



Association of physical activity and hydration markers with cellular health in older adults: A doubly labeled water study

Kan Oishi^{a,b,c}, Tsukasa Yoshida^{c,d,e}, Hyeon-Ki Kim^{c,e}, Keiichi Yokoyama^f, Yuya Watanabe^{d,g}, Yui Nakayama^h, Misaka Kimura^d, Kojiro Ishii^b, Rei Ono^c, Yosuke Yamada^{c,d,e,h,*}

^a Faculty of Education, Saga University, 1 Honjo-machi, Saga, Saga 840-8502, Japan

^b Faculty of Health and Sports Science, Doshisha University, 1-3 Tatara Miyakodani, Kyotanabe, Kyoto 610-0394, Japan

^c Center for Physical Activity Research, National Institutes of Biomedical Innovation, Health, and Nutrition, Japan Kento Innovation Park NK Building, 3-17 Senrioka Shinmachi, Settsu, Osaka 566-0002, Japan

^d Institute of Interdisciplinary Research, Institute for Active Health, Kyoto University of Advanced Science, 1-1 Nanjo Otani, Sogabe-cho, Kameoka, Kyoto 621-8555, Japan

^e Graduate School of Medicine, Tohoku University, 2-1 Seiryomachi Aoba-ku, Sendai, Miyagi 980-8575, Japan

^f Non-Profit Organization Genki-Up AGE Project, 1-46 Umahori Mukaibata, Shino-cho, Kameoka, Kyoto 621-0823, Japan.

^g Faculty of Sport Study, Biwako Seikei Sport College, 1,204 Kitahira, Otsu, Shiga 520-0503, Japan

^h Graduate School of Biomedical Engineering, Tohoku University, 6-6-12 Aramaki Aza Aoba Aoba-ku, Sendai, Miyagi 980-8579, Japan

ARTICLE INFO

Section Editor: Christiaan Leeuwenburgh

Keywords:

Phase angle
Extracellular water-to-intracellular water ratio
Physical activity
Doubly labeled water
Older adults

ABSTRACT

Purpose: This study aimed to determine the associations between the phase angle (PhA), extracellular water-to-intracellular water ratio (ECW/ICW), physical activity levels (PALs) measured using the doubly labeled water (DLW) method, and the time spent engaging in physical activities of varying intensities in older adults living in Japan.

Methods: This study was part of the Kyoto–Kameoka prospective cohort study of older adults. The PhA and ECW/ICW ratio were measured using a multi-frequency bioelectrical impedance analysis device. PAL, percentage body fat, and fat-free mass index were calculated using the DLW method, while the average daily moderate-to-vigorous physical activity (MVPA), high-intensity light physical activity (HLPa), low-intensity light physical activity, and sedentary behavior durations were quantified using a triaxial accelerometer.

Results: Adjusted partial order correlation analysis showed significant positive correlations between the PhA and PAL ($p = 0.001$), as well as between the PhA and the MVPA ($p = 0.027$) and HLPa ($p = 0.023$) durations. The PAL was also significantly negatively correlated with the ECW/ICW ratio ($p = 0.008$). Adjusted multiple regression analysis confirmed that PAL ($p = 0.004$) and HLPa ($p = 0.025$) were significantly positively associated with PhA and negatively associated with ECW/ICW ratio ($p = 0.004$ and $p = 0.036$, respectively).

Conclusions: PAL, measured using the precise DLW method, is an explanatory factor for PhA and ECW/ICW ratio. HLPa below the MVPA threshold was linked to more favorable hydration markers of cellular health in older adults. As this was a cross-sectional study, causality cannot be inferred.

1. Introduction

Bioelectrical impedance analysis (BIA) uses a weak alternating current applied to the human body to measure its internal resistance (impedance), enabling a non-invasive and simple estimation of body composition with high accuracy and reproducibility (Baumgartner et al., 1988). Its resistive component can be separated into resistance and reactance (Baumgartner et al., 1988). Resistance represents the voltage

drop, indicating the conductivity of ionic solutions in body fluids (water), while reactance represents the current flow delay, measured as a phase shift, reflecting the capacitance of cell membranes and tissue interfaces (Norman et al., 2012). Recently, the use of raw impedance values in determining nutritional status has garnered significant attention (Lukaski et al., 2017; Kuchnia et al., 2017).

The phase angle (PhA) is defined as the arctangent of reactance divided by resistance (Lukaski, 1996). It is calculated as the angle of cell

* Corresponding author at: Graduate School of Biomedical Engineering, Tohoku University, 2-1 Seiryomachi, Aoba-ku, Sendai, Miyagi 980-8575, Japan.
E-mail address: yosuke.yamada.c1@tohoku.ac.jp (Y. Yamada).

<https://doi.org/10.1016/j.exger.2025.112859>

Received 1 July 2025; Received in revised form 31 July 2025; Accepted 6 August 2025

Available online 11 August 2025

0531-5565/© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

membrane resistance and serves as an indicator of nutritional status, aging, and cellular and cell membrane integrity (Norman et al., 2012; Lukaski, 1996). Based on theoretical principles, multi-frequency BIA can estimate body water content by dividing it into extracellular water (ECW) and intracellular water (ICW) (Yamada et al., 2010). Skeletal muscle, which contains a significant amount of water, uses myocyte membranes to separate the ICW and ECW (Yamada et al., 2010). ICW reflects myocyte mass, while ECW represents the sum of interstitial fluid and plasma in the extracellular space (Yamada et al., 2010); therefore, a decrease in the ECW/ICW ratio is considered indicative of a relative increase in contractile tissue, primarily skeletal muscle fibers. The PhA has been reported to be associated with grip strength and upper limb muscle mass (Akamatsu et al., 2022), and a low PhA has been suggested to increase the risk of adverse events and mortality (Garlini et al., 2019; Ceolin et al., 2023). Similarly, the ECW/ICW ratio has been associated with T2 relaxation time and muscle volume measured by magnetic resonance imaging, suggesting its potential as an indicator of physical function and prognoses (Yamada et al., 2017a; Yamada et al., 2023; Asano et al., 2023a; da Costa Pedro et al., 2024). These findings highlight PhA and ECW/ICW ratio as independent determinants of physical performance and may serve as indicators of muscle quality (Akamatsu et al., 2022; Yamada et al., 2017a; Iwasaka et al., 2023; Costa Pereira et al., 2024).

Physical activity has recently been proposed as a possible explanatory factor for the PhA and ECW/ICW ratio (Iwasaka et al., 2023; Yamada et al., 2022). Utilizing a triaxial accelerometer, Yamada et al. (Yamada et al., 2022) found that moderate-to-vigorous physical activity (MVPA), step count, and subjective exercise habits were positively associated with the PhA in middle-aged to older adults in Japan. Additionally, Iwasaka et al. (Iwasaka et al., 2023) found that the physical activity level (PAL), which refers to the average activity intensity, was associated with the ECW/ICW ratio in middle-aged and older adults in Japan. However, the relationship between physical activity, PhA, and the ECW/ICW ratio remains poorly understood.

To measure physical activity, PAL is calculated as the total energy expenditure (TEE) divided by the basal metabolic rate (BMR). The gold standard for measuring TEE under free-living conditions is the doubly labeled water (DLW) method (Westertorp, 1999); however, previous studies examining the relationship between PAL and PhA or ECW/ICW ratio used a triaxial accelerometer to assess physical activity (Iwasaka et al., 2023; Yamada et al., 2022). Notably, TEE values calculated using triaxial accelerometers can be underestimated in older adults (Yamada et al., 2018). In addition, a previous study measuring the TEE and PAL using both triaxial accelerometers and the DLW method in patients with chronic obstructive pulmonary disease (COPD)—using Bland–Altman analysis to evaluate agreement—found significant proportional and additive errors (Sato et al., 2021). These discrepancies suggest that PAL values derived from triaxial accelerometers may lack absolute and relative validity, with error differences varying by individual.

Previous studies have mainly examined the relationship between the PhA or ECW/ICW ratio and physical activity intensity in middle-aged and older adults (Iwasaka et al., 2023; Yamada et al., 2022). However, few significant insights regarding the relationship between physical activity and the PhA or ECW/ICW ratio in older adults have been revealed. Given that the PhA decreases while the ECW/ICW ratio increases with age (Ohashi et al., 2018; Mattiello et al., 2020), understanding these relationships is crucial. A previous systematic review showed that low-intensity physical activity is beneficial for certain health outcomes in older adults (Amagasa et al., 2018). Additionally, the relationships between different intensities of physical activity and various health indicators in older adults have been examined by further classifying light physical activity (LPA) into high-intensity light physical activity (HLLPA) and low-intensity light physical activity (LLPA). For example, a previous study reported that HLLPA, but not LLPA, was significantly associated with physical function in women (Izawa et al., 2017). This suggests that the relationship between intensity-based

physical activity and the PhA or ECW/ICW ratio in older adults may differ from that observed in other age groups. A more detailed classification of LPA could provide valuable insights into its role in maintaining the PhA and ECW/ICW ratio in older adults.

Given this context, we aimed to determine the relationship between the PhA, ECW/ICW ratio, and PAL measured using the more accurate DLW method, as well as the durations of physical activity at different intensities measured using triaxial accelerometers in older adults.

2. Methods

2.1. Study design and participants

This study was conducted using a cross-sectional design and formed part of the Kyoto–Kameoka study, a community-based prospective cohort study of older adults initiated on July 29, 2011. The Kyoto–Kameoka study included 13,294 residents of Kameoka City, Kyoto Prefecture, Japan. Details of this cohort have been described elsewhere (Yamada et al., 2017b). A needs assessment survey was conducted in the daily living area by mail, followed by a health and nutritional status survey in February 2012. Approximately 13,294 responses were collected between July 29th 2011, and February 13th 2012. However, individuals certified as requiring support (level 1 or 2) or nursing care (level 1 or 2) ($n = 1240$) and those who had died ($n = 69$) were excluded, leaving a total of 11,985 respondents. Of these, 8370 completed the additional health and nutrition survey. After removing 30 respondents with unidentifiable information and 21 whose sex differed between the first and additional surveys, 8319 valid responses remained. Approximately 1379 of the 8319 participants who had been physically examined were informed of the current study plan and invited to participate in May 2012. The average temperature and humidity in the region during the study period were 18.8 °C and 60 %, respectively. Ultimately, 175 volunteers provided informed consent, and 144 completed the DLW study. After excluding ten participants with missing items for analysis, 134 (62 women, 72 men) individuals were included in the final analysis.

2.2. Ethical statement

This study was approved by the Ethics Committee of the Kyoto Prefectural University of Medicine (approval number: RBMR-E-363) and the National Institute of Health and Nutrition (NIHN187-3) and was conducted in compliance with the Declaration of Helsinki. All participants were informed of the experimental details and provided informed consent.

2.3. Anthropometric measurements and bioelectrical impedance method

Participants' height and weight were measured to the nearest 0.1 cm and 0.1 kg, respectively, using a stadiometer and digital scale (DST-210S; Muratec-KDS Corp., Kyoto, Japan). Using these values, the body mass index (BMI) was calculated. Body composition was assessed using a multi-frequency BIA device (IMP SFB7; ImpediMed, Brisbane, Qld, Australia) to obtain impedance data at frequencies of 5, 50, and 250 kHz. Electrodes (Red Dot; 3M, Tokyo, Japan) were placed between the second and third metacarpals and metatarsals to measure current flow and between the radius and ulna and medial and lateral ankles to measure voltage. The PhA was calculated from the impedance data at 50 kHz, while the ECW/ICW ratio was calculated from the impedance data at 5 and 250 kHz.

2.4. Measurement of TEE and calculation of body composition and PALs using the DLW method

PALs were measured over 14 days using the DLW method, which is considered more accurate than an activity meter (Yamada et al., 2018). The DLW measurement sequence was as follows:

First, urine samples were collected from participants arriving at the study site on day 0, and the baseline levels of ^2H and ^{18}O accumulation were measured. Subsequently, for dose administration, participants received a premixed drink containing approximately 0.12 g/kg estimated total body water (TBW) of $^2\text{H}_2\text{O}$ (99.8 at.%, Taiyo Nippon Sanso, Tokyo, Japan) and 2.5 g/kg estimated TBW of H_2^{18}O (10.0 at.%, Taiyo Nippon Sanso, Tokyo, Japan) between 7:00 and 8:00 a.m. Follow-up urine samples were then collected first thing in the morning on the following day (day 1), day 2, day 8, day 9, day 15, and day 16 after dose administration. Urine samples were stored at -15°C for subsequent analysis using an isotope ratio mass spectrometer (Hydra 20-20, SerCon Ltd., Crewe, UK).

Upon analysis, CO_2 and platinum-catalyzed H_2 were used to equilibrate the ^{18}O - and ^2H , respectively. The intercept method was used to determine the dilution spaces of ^{18}O (No) and ^2H (Nd) by dividing them according to the administered tracer dose (Yamada et al., 2018). Since all ^{18}O enrichment values in urine samples on day 16 were above 8 ‰, the intercept method was used to analyze urine samples collected at baseline and on days 1, 2, 8, 9, 15, and 16. The dilution spaces of Nd and No indicate the distribution or dispersion of isotopes (^{18}O and ^2H) within the body's water compartment. These isotope dilution spaces are close to but slightly larger than that of the TBW because both ^{18}O and ^2H are also exchanged with non-aqueous components in the body. The estimated extent of non-aqueous exchange is approximately 1 ‰ and 4–5 ‰ for ^{18}O and ^2H , respectively, in humans (Haggarty et al., 1988; Racette et al., 1994). Consequently, the relative abundance ratio influenced the TBW value determined using the DLW method. The Nd/No values in this study (1.033 ± 0.013 ; range, 0.999–1.077) were comparable to those reported in a previous large-scale study (Sagayama et al., 2016). Therefore, TBW (mol) was calculated as the average of Nd (mol) divided by 1.043 and No (mol) divided by 1.007 (Speakman et al., 2021).

The carbon dioxide production rate (rCO_2 , mol/d) was calculated using the following formula:

$$\text{rCO}_2 = 0.4554 \times \text{TBW} \times (1.007 k_o - 1.043 k_d)$$

In this formula, k_o and k_d represent the daily excretion rates of ^{18}O and ^2H , respectively.

Based on the calculated rCO_2 (mol/d) and estimated 24-hour respiratory exchange ratio (RER), TEE was then determined using the modified Wier formula:

$$\text{TEE (kcal/d)} = 22.6 \times \left(3.94 \times \left(\frac{\text{rCO}_2}{\text{RER}} \right) + 1.106 \times \text{rCO}_2 \right)$$

In this formula, 22.6 (L/mol) is the molar volume of carbon dioxide (Weir, 1949).

A detailed DLW method quality control checklist is provided in the International Atomic Energy Agency (IAEA) document (International Atomic Energy Agency, 2009). The fat-free mass (FFM; kg) was calculated from the TBW using the adult hydration factor (0.732), as described in a separate IAEA document (International Atomic Energy Agency, 2011). Subsequently, the fat mass (FM; kg) was calculated by subtracting the FFM from the body weight. The percentage of body fat (PBF; %) was then calculated as follows:

$$\text{PBF (\%)} = \frac{\text{FM (kg)}}{\text{body weight (kg)}} \times 100$$

Furthermore, the fat-free mass index (FFMI; kg/m^2) was calculated using the following formula:

$$\text{FFMI (kg/m}^2\text{)} = \frac{\text{FFM (kg)}}{\text{height (m}^2\text{)}}$$

Next, we estimated the BMR using the following formula described by Gampule et al. (Ganpule et al., 2007):

$$\begin{aligned} \text{BMR (kcal)} = & (2.3958 + 0.0787 \times \text{FFM} - 0.0109 + 0.0268 \\ & \times \text{FM} - 0.3314 \times \text{sex (Male : 0, Female : 1)}) \end{aligned}$$

Finally, the PAL was calculated by dividing the TEE by the BMR.

2.5. Measurement of the intensity-specific physical activity time using an accelerometer

All participants wore a triaxial accelerometer (Actimarker; Panasonic, Osaka, Japan) on their waist during the DLW measurements, except when bathing and sleeping. Using this data, the average daily MVPA, HLPa, LLPa, and sedentary behavior (SB) durations were assessed. These were defined as follows: MVPA, $x \geq 3$ metabolic equivalents of task (METs)/min; HLPa, $2.0 \leq x < 3.0$ METs/min; LLPa, $1.5 \leq x < 2.0$ METs/min; and SB, $x < 1.5$ METs/min. These definitions were based on previous studies (Izawa et al., 2017).

2.6. Other variables

In this study, we investigated energy intake (EI) and protein intake (PI) using a Japanese food frequency questionnaire (FFQ). EI was then calibrated utilizing a previously developed equation that uses TEE measured using the DLW method (Watanabe et al., 2019a). The equation used for this calculation attenuates the impact of age, sex, and BMI-related self-reporting bias on estimated EI in older adults, assessed via the FFQ (Watanabe et al., 2022). We calculated EI based on ideal body weight (IBW) (kcal/day/kgIBW) by dividing EI by the IBW. Based on a previous study in Japanese individuals (Watanabe et al., 2023), IBW was calculated as $22 \times \text{height squared (m}^2\text{)}$. Since EI per IBW (EI/IBW) has been shown to be associated with mortality, the presence or absence of chronic diseases was also assessed using a questionnaire. PI was also calculated using an equation previously validated in studies involving older adults with dietary records (Watanabe et al., 2019b).

2.7. Statistical analysis

Continuous variables are expressed as the mean and standard deviation (SD) or median and interquartile range. All variables were compared according to sex or chronic disease status, with non-normally distributed continuous variables analyzed using the Mann–Whitney *U* test, normally distributed continuous variables analyzed using the unpaired *t*-test, and nominal variables analyzed using the chi-square test. Partial order correlation analyses were conducted to determine the relationships between age and BMI, FFMI, PBF, EI/IBW, PhA, ECW/ICW, PI, PAL, MVPA, HLPa, LLPa, and SB, adjusted for sex and chronic disease status, and between the PhA or ECW/ICW ratio and PAL, MVPA, HLPa, LLPa, and SB, adjusted for age, sex, chronic disease status, FFMI, PBF, EI/IBW, and PI.

Additionally, we examined the independent association between the PhA or ECW/ICW ratio and each physical activity index using multiple regression analysis with PhA or the ECW/ICW ratio as objective variables; PAL, MVPA, HLPa, LLPa, and SB as explanatory variables; and age, sex, chronic disease status, FFMI, PBF, EI/IBW, and PI as covariates. The multiple regression analysis was divided into two models, with Model 1 including only the PAL as the explanatory variable and Model 2 including the duration of physical activity at all intensities as simultaneous explanatory variables. Finally, multiple regression analysis was conducted with the PAL as the objective variable, physical activity duration at each intensity as the explanatory variables, and age, sex, chronic disease status, FFMI, PBF, EI/IBW, PI as covariates.

All statistical analyses were performed using SPSS version 30 (IBM Corp., Armonk, NY), with statistical significance set at $p < 0.05$.

3. Results

3.1. Participant characteristics

Table 1 presents the mean and SD of age, height, weight, BMI, FFM, FFMI, FM, PBF, EI, IBW, EI/IBW, PI, PhA, ECW/ICW, PAL, MVPA, HLPa, LLPA, and SB, as well as the results of the comparisons between men and women or between individuals with and without chronic disease. Notably, men had a significantly higher height ($p < 0.001$), weight ($p < 0.001$), FFM ($p < 0.001$), FFMI ($p < 0.001$), EI ($p < 0.001$), IBW ($p < 0.001$), PI ($p = 0.001$), PhA ($p < 0.001$), and ECW/ICW ($p < 0.001$) than women. Conversely, women had a significantly higher PBF ($p < 0.001$), PAL ($p = 0.026$), MVPA ($p = 0.020$), HLPa ($p < 0.001$), LLPA ($p < 0.001$), and SB ($p = 0.001$) than men. No significant differences were observed in any of the variables compared between individuals with and without chronic disease.

3.2. Partial order correlation analysis

Table 2 displays the partial order correlation coefficients between age, BMI, FFM, PBF, EI/IBW, or PI and the PhA, ECW/ICW ratio, or each physical activity indicator adjusted for sex and chronic disease status. Notably, the ECW/ICW ratio showed a significant positive correlation with age ($r = 0.304$; $p < 0.001$), while the PhA ($r = -0.353$; $p < 0.001$) and MVPA ($r = -0.332$; $p < 0.001$) showed significant negative correlations with age. Moreover, the PhA showed a significant positive correlation with BMI ($r = 0.175$; $p = 0.045$), while LLPA showed a significant negative correlation with BMI ($r = -0.191$; $p = 0.028$). The PhA also showed a significant positive correlation with FFMI ($r = 0.240$; $p = 0.006$), while the ECW/ICW ratio showed a significant negative correlation with FFMI ($r = -0.288$; $p < 0.001$). Furthermore, the PAL ($r = -0.264$; $p = 0.002$), HLPa ($r = -0.182$; $p = 0.037$), and LLPA ($r = -0.280$; $p = 0.001$) showed significant negative correlations with PBF, while the PhA showed a significant positive correlation with EI/IBW (r

$= 0.211$; $p = 0.015$).

Table 3 shows the partial order correlation coefficients between the PhA, ECW/ICW ratio, and each physical activity indicator, adjusted for sex, chronic disease status, age, FFM, PBF, EI/IBW, and PI. Notably, the PAL ($r = 0.286$; $p = 0.001$), MVPA ($r = 0.197$; $p = 0.027$), and HLPa ($r = 0.201$; $p = 0.023$) showed significant positive correlations with the PhA. In contrast, the PAL ($r = -0.230$; $p = 0.009$) was significantly negatively correlated with the ECW/ICW ratio.

3.3. Multiple linear regression analysis

Table 4 displays the results of multiple regression analysis with the PhA as the objective variable; each physical activity indicator (PAL, MVPA, HLPa, LLPA, and SB) as an explanatory variable; and sex, chronic disease status, age, FFM, PBF, EI/IBW, and PI as covariates. In Model 1, the PAL was significantly positively associated with the PhA ($\beta = 0.200$; $p = 0.005$). In Model 2, where all intensity-specific physical activity durations were included as explanatory variables, only HLPa showed a significant positive association with the PhA ($\beta = 0.251$; $p = 0.026$). A model including interaction terms between sex and PAL or HLPa as explanatory variables is presented in Supplementary Material 1.

Table 5 presents the results of regression analysis using the ECW/ICW ratio as the objective variable. In Model 1, PAL showed a significant negative association with the ECW/ICW ratio ($\beta = -0.189$; $p = 0.004$). In Model 2, only HLPa showed a significant negative association with the ECW/ICW ratio ($\beta = -0.217$; $p = 0.037$). A model including interaction terms between sex and PAL or HLPa as explanatory variables is presented in Supplementary Material 2.

Table 6 shows the results of regression analysis using the PAL as the objective variable; all intensity-specific physical activity durations as explanatory variables; and sex, chronic disease status, age, FFM, PBF, EI/IBW, and PI as covariates. Notably, both MVPA ($\beta = 0.287$; $p = 0.001$) and HLPa ($\beta = 0.392$; $p < 0.001$) showed significant positive

Table 1
Patient characteristics.

Characteristics	Overall (n = 134)	Men (n = 72)	Women (n = 62)	p^a	Chronic diseases (n = 47)	Non-chronic diseases (n = 87)	p^b
Age (years) ^d	73 (69–77)	74 (69–78)	72 (69–76)	0.331	73 (69–77)	73 (69–77)	0.881
Height (cm) ^d	159.5 (150.8–164.1)	163.8 (161.0–167.7)	150.0 (147.0–154.0)	<0.001	159.9 (152.7–163.3)	158.7 (149.8–164.4)	0.606
Weight (kg) ^c	56.6 ± 9.8	61.6 ± 8.9	50.8 ± 7.4	<0.001	57.1 ± 9.9	56.4 ± 9.8	0.678
BMI (kg/m ²) ^c	22.6 ± 3.2	22.8 ± 2.9	22.4 ± 3.4	0.430	22.7 ± 3.2	22.6 ± 3.1	0.911
FFM (kg) ^d	39.1 (32.5–44.0)	43.7 (41.4–47.7)	32.2 (30.6–35.3)	<0.001	39.8 (32.8–43.9)	37.8 (32.5–44.8)	0.762
FFMI (kg/m ²) ^c	15.5 ± 1.8	16.4 ± 1.6	14.4 ± 1.5	<0.001	15.5 ± 1.8	15.5 ± 1.9	0.953
FM (kg) ^c	17.7 ± 5.3	17.3 ± 5.2	18.1 ± 5.5	0.365	18.0 ± 5.8	17.5 ± 5.0	0.629
PBF (%) ^c	31.0 ± 3.8	27.6 ± 5.5	35.0 ± 6.9	<0.001	31.1 ± 7.8	31.0 ± 6.9	0.926
EI (kcal/day) ^d	2173 (1904–2353)	2340 (2269–2425)	1884 (1780–2024)	<0.001	2193 (1959–2367)	2130 (1891–2351)	0.606
IBW (kg) ^d	55.9 (50.0–59.2)	59.0 (57.0–61.9)	49.5 (47.5–52.2)	<0.001	56.2 (51.3–58.7)	55.4 (49.4–59.5)	0.606
EI/IBW (kcal/day/kg) ^c	39.0 ± 3.8	39.6 ± 3.0	38.3 ± 4.5	0.055	39.0 ± 3.6	39.0 ± 4.0	0.924
PI (kg/day) ^d	58.4 (50.7–66.8)	61.6 (53.7–70.7)	55.8 (45.3–62.5)	0.001	57.8 (50.3–66.3)	59.6 (51.3–67.7)	0.574
PhA (degree) ^c	5.1 ± 0.7	5.4 ± 0.6	4.7 ± 0.5	<0.001	5.0 ± 0.7	5.1 ± 0.7	0.476
ECW/ICW ^c	0.85 ± 0.06	0.89 ± 0.06	0.81 ± 0.04	<0.001	0.86 ± 0.06	0.85 ± 0.07	0.306
PAL ^d	1.77 (1.64–1.91)	1.73 (1.59–1.87)	1.80 (1.68–1.96)	0.026	1.72 (1.57–1.95)	1.78 (1.66–1.91)	0.106
MVPA (min/day) ^d	37 (21–59)	31 (15–50)	42 (32–63)	0.020	35 (21–49)	39 (20–62)	0.249
HLPa (min/day) ^d	195 (143–242)	170 (119–216)	223 (172–252)	<0.001	188 (131–221)	212 (143–245)	0.144
LLPA (min/day) ^d	181 (150–222)	161 (128–197)	210 (180–238)	<0.001	181 (147–222)	182 (154–224)	0.702
SB (min/day) ^d	186 (152–225)	165 (135–206)	201 (165–249)	0.001	196 (147–229)	181 (152–225)	0.565

Data are presented as the mean ± SD or median (interquartile range).

SD, standard deviation; BMI, body mass index; FFM, fat-free mass; FFMI, fat-free mass index; FM, fat mass; PBF, percent body fat; EI, energy intake; IBW, ideal body weight; EI/IBW, energy intake per ideal body weight; PI, protein intake; PhA, phase angle; ECW/ICW, extracellular water-to-intracellular water ratio; PAL, physical activity level; MVPA, moderate-to-vigorous physical activity; HLPa, high-intensity light physical activity; LLPA, low-intensity light physical activity; SB, sedentary behavior; Bold font denotes $p < 0.05$.

^a Comparison by sex.

^b Comparison by presence or absence of chronic diseases.

^c Expressed as mean ± standard deviation due to a normal distribution.

^d Expressed as median (interquartile range) due to the absence of a normal distribution.

Table 2

Partial order correlation analysis examining the relationship between age, BMI, FFMI, PBF, or EI/IBW and the PhA, ECW/ICW ratio, or each physical activity indicator^a.

Variables		PhA (degree)	ECW/ICW	PAL	MVPA (min/day)	HLPa (min/day)	LLPA (min/day)	SB (min/day)
Age (years)	<i>r</i>	-0.353	0.304	-0.041	-0.332	0.051	0.017	-0.124
	<i>p</i>	<0.001	<0.001	0.638	<0.001	0.562	0.850	0.158
BMI (kg/m ²)	<i>r</i>	0.175	-0.167	-0.151	-0.050	-0.155	-0.191	-0.076
	<i>p</i>	0.045	0.055	0.085	0.570	0.076	0.028	0.389
FFMI (kg/m ²)	<i>r</i>	0.240	-0.288	0.034	0.017	-0.043	-0.043	-0.011
	<i>p</i>	0.006	<0.001	0.700	0.848	0.626	0.625	0.902
PBF (%)	<i>r</i>	0.015	0.056	-0.264	-0.124	-0.182	-0.280	-0.139
	<i>p</i>	0.863	0.522	0.002	0.156	0.037	0.001	0.113
EI/IBW (kcal/day/kg)	<i>r</i>	0.211	-0.170	-0.017	-0.059	0.051	0.086	-0.033
	<i>p</i>	0.015	0.051	0.845	0.502	0.564	0.326	0.709
PI (kg/day)	<i>r</i>	0.020	0.001	-0.105	-0.088	-0.002	-0.049	-0.076
	<i>p</i>	0.817	0.990	0.231	0.314	0.985	0.574	0.387

BMI, body mass index; FFMI, fat-free mass index; PBF, percent body fat; EI/IBW, energy intake per ideal body weight; PI, protein intake; PhA, phase angle; ECW/ICW, extracellular water-to-intracellular water ratio; PAL, physical activity level; MVPA, moderate-to-vigorous physical activity; HLPa, high-intensity light physical activity; LLPA, low-intensity light physical activity; SB, sedentary behavior; *r*, partial correlation coefficient; Bold font denotes $p < 0.05$.

^a Adjusted for sex and chronic disease status.

Table 3

Partial order correlation analysis examining the relationship between the PhA or ECW/ICW ratio and each varying intensity of physical activity^a.

Variables		PAL	MVPA (min/day)	HLPa (min/day)	LLPA (min/day)	SB (min/day)
PhA (degree)	<i>r</i>	0.286	0.197	0.201	0.029	-0.069
	<i>p</i>	0.001	0.027	0.023	0.745	0.440
ECW/ICW	<i>r</i>	-0.230	-0.165	-0.146	-0.064	-0.033
	<i>p</i>	0.009	0.064	0.101	0.477	0.709

PAL, physical activity level; MVPA, moderate-to-vigorous physical activity; HLPa, high-intensity light physical activity; LLPA, low-intensity light physical activity; SB, sedentary behavior; PhA, phase angle; ECW/ICW, extracellular water-to-intracellular water ratio; *r*, partial correlation coefficient; FFMI, fat-free mass index; PBF, percent body fat; EI/IBW, energy intake per ideal body weight; PI, protein intake; Bold font denotes $p < 0.05$.

^a Adjusted for sex, chronic disease status, age, FFMI, PBF, EI/IBW, and PI.

Table 4

Multiple regression analysis examining the relationship between the PhA and physical activity^a.

Explanatory variable	Model 1 [†]				Model 2 [‡]			
	B	SE	β	<i>p</i>	B	SE	β	<i>p</i>
PAL	0.478	0.166	0.200	0.005				
MVPA (10 min/day)					0.008	0.018	0.036	0.673
HLPa (10 min/day)					0.023	0.010	0.251	0.026
LLPA (10 min/day)					-0.022	0.014	-0.186	0.102
SB (10 min/day)					0.016	0.011	0.130	0.147

PhA, phase angle; PAL, physical activity level; MVPA, moderate-to-vigorous physical activity; HLPa, high-intensity light physical activity; LLPA, low-intensity light physical activity; SB, sedentary behavior; B, partial regression coefficient; SE, standard error; β , standardized partial regression coefficient; FFMI, fat-free mass index; PBF, percent body fat; EI/IBW, energy intake per ideal body weight; PI, protein intake; Bold font denotes $p < 0.05$.

^a Adjusted for sex, chronic disease status, age, FFMI, PBF, EI/IBW, and PI.

[†] $R = 0.689$, adjusted $R^2 = 0.474$, $F = 14.093$, $p < 0.001$.

[‡] $R = 0.691$, adjusted $R^2 = 0.431$, $F = 10.143$, $p < 0.001$.

Table 5

Multiple regression analysis examining the relationship between the ECW/ICW ratio and varying intensities of physical activity^a.

Explanatory variable	Model 1 [†]				Model 2 [‡]			
	B	SE	β	<i>p</i>	B	SE	β	<i>p</i>
PAL	-0.044	0.015	-0.189	0.004				
MVPA (10 min/day)					-0.001	0.002	-0.045	0.574
HLPa (10 min/day)					-0.002	0.001	-0.217	0.037
LLPA (10 min/day)					0.002	0.001	0.143	0.173
SB (10 min/day)					-0.001	0.001	-0.118	0.157

ECW/ICW, extracellular water-to-intracellular water ratio; PAL, physical activity level; MVPA, moderate-to-vigorous physical activity; HLPa, high-intensity light physical activity; LLPA, low-intensity light physical activity; SB, sedentary behavior; B, partial regression coefficient; SE, standard error; β , standardized partial regression coefficient; FFMI, fat-free mass index; PBF, percent body fat; EI/IBW, energy intake per ideal body weight; PI, protein intake; Bold font denotes $p < 0.05$.

^a Adjusted for sex, presence of chronic diseases, age, FFMI, PBF, EI/IBW, and PI.

[†] $R = 0.740$, adjusted $R^2 = 0.519$, $F = 18.915$, $p < 0.001$.

[‡] $R = 0.741$, adjusted $R^2 = 0.508$, $F = 13.468$, $p < 0.001$.

Table 6

Multiple regression analysis examining the relationship between the PAL and varying intensities of physical activity^a.

Explanatory variable	B	SE	β	p
MVPA (10 min/day)	0.0251	0.0077	0.2871	0.001
HLPa (10 min/day)	0.0151	0.0044	0.3918	<0.001
LLPA (10 min/day)	0.0062	0.0058	0.1220	0.290
SB (10 min/day)	−0.0002	0.0047	0.0042	0.963

The statistical results for this model were as follows: $R = 0.674$, adjusted $R^2 = 0.405$, $F = 9.235$, $p < 0.001$.

PAL, physical activity level; MVPA, moderate-to-vigorous physical activity; HLPa, high-intensity light physical activity; LLPA, low-intensity light physical activity; SB, sedentary behavior; B, partial regression coefficient; SE, standard error; β , standardized partial regression coefficient; FFMI, fat-free mass index; PBF, percent body fat; EI/IBW, energy intake per ideal body weight; Bold font denotes $p < 0.05$.

^a Adjusted for sex, presence of chronic diseases, age, FFMI, PBF, EI/IBW, and PI.

associations with the PAL.

4. Discussion

To the best of our knowledge, this study is the first to report an association between the PhA or ECW/ICW ratio, PAL measured using the DLW method, and intensity-detailed physical activity in older adults. Our findings indicate that the PAL and HLPa are significantly associated with the PhA and ECW/ICW ratio, even after adjusting for confounding factors in this population. These findings suggest that the PAL and HLPa, which are closely related to PAL, may be important correlates of the PhA and ECW/ICW ratio in older adults.

We observed that the PhA and ECW/ICW ratio were significantly associated with the PAL even after adjusting for sex, chronic disease status, age, FFMI, PBF, EI/IBW, and PI. This finding supports the results of previous research on middle- to older-aged participants in which PALs were measured using a triaxial accelerometer (Iwasaka et al., 2023; Yamada et al., 2022). In this study, we also evaluated PALs using the DLW method, which provides more accurate assessment of physical activity than triaxial accelerometers and is considered the gold standard assessment method under free-living conditions. Our finding that PAL measured using the DLW method was significantly associated with PhA strengthens the existing evidence linking PALs measured by triaxial accelerometers to the PhA and ECW/ICW ratio.

Additionally, we examined the relationship between the PhA or ECW/ICW ratio and the time spent on physical activities of different intensities, measured by an accelerometer, in older adults. We found that when the duration of physical activity at each intensity was analyzed separately, the durations of HLPa and MVPA were significantly related to the PhA. This finding align with previous studies reporting that high-intensity physical activities, such as MVPA, contributes to then improvement in PhA (Yamada et al., 2022; Asano et al., 2023b; Nakashima et al., 2024). The significant relationship between HLPa and PhA may be partially explained by the change in relative activity intensity, as indicated by absolute activity intensity, with age. A previous study reported that daily use of approximately 20–30 % of maximal muscle strength is required to maintain high muscle strength (Müller, 1970). This threshold is classified as MVPA in the general adult population. According to the American College of Sports Medicine Position Stand, METs measure absolute exercise intensity; however, the relative intensity indicated by METs varies with age (Pollock et al., 1998). An exercise intensity of 2–3 METs is considered ‘very light’ to ‘light’ for people in their 20s–30s, but ‘moderate’ for those in their 80s (Pollock et al., 1998). Therefore, in older adults, HLPa—which is less intense than MVPA—may still represent more than 20 % of maximal

strength. Notably, Osuka et al. (Osuka et al., 2015) have previously shown that muscle strength is maintained with LPA in older Japanese participants. Moreover, LPA has been shown to be significantly positively associated with muscle strength (grip strength) in older adults (Ramsey et al., 2021). Our findings suggest that HLPa, which is defined as 2–3 METs/min and is lower in intensity than MVPA, may contribute to the maintenance and improvement of the PhA and muscle strength in older adults.

In Model 2 of the multiple regression analysis, which included all intensity-specific physical activity durations as explanatory variables, only HLPa was significantly associated with the PhA and ECW/ICW ratio. While definitive explanations are lacking, this finding may reflect the specific characteristics of the study participants. The pattern of intensity-specific physical activity in older adults differs from that in younger and middle-aged individuals; with advancing age, LPA tends to increase while MVPA decreases (Ayabe et al., 2009). In this study, the mean durations of LPA and MVPA were 384.0 ± 113.3 min and 43.2 ± 31.9 min, respectively. When compared with the findings of a previous study on middle-aged and older adults in Japan (Yamada et al., 2022), participants in this study spent more time on LPA and less time on MVPA. Furthermore, within the LPA category, the participants in this study spent more time on HLPa than on LLPA. Multiple regression analysis conducted with PAL (indicating average exercise intensity) as the dependent variable; times spent on physical activity by intensity as explanatory variables; and sex, chronic disease status, age, FFMI, PBF, EI/IBW, and PI as covariates revealed significant positive associations for both MVPA and HLPa, with the standardized partial regression coefficient being greater for HLPa. Additionally, HLPa values spanned the widest range across the physical activities of each intensity in this study. This suggests that, among the participants included in this study, HLPa had a more significant effect on the PhA and ECW/ICW ratio than MVPA, likely due to the longer time spent on HLPa, its stronger effect on average exercise intensity, and its broader distribution, which facilitated the identification of a significant relationship.

Despite its various strengths, this study had some limitations. First, its cross-sectional design precludes the ability to infer causal relationships. A previous longitudinal study demonstrated that PhA may predict mobility limitations in older men (Asano et al., 2023c), suggesting the possibility of reverse causality in the present findings. Therefore, further longitudinal research is warranted to clarify the causal relationship. Second, the sample size limited the ability to conduct stratified analyses according to sex or age. Nonetheless, we tested models that included interaction terms between sex and PAL or HLPa as explanatory variables; however, neither interaction was significantly associated with the outcomes. Additionally, using G*Power 3.1.9.2 (Faul et al., 2009), the required sample size was calculated to be 123 based on an effect size (f^2) of 0.15, α error probability of 0.05, power $(1-\beta)$ of 0.8, and 11 predictors. Therefore, the sample size of this study was sufficient to conduct an overall analysis. Third, as the participants in this study were older adults living in a single region of Japan, the generalizability of the findings to older adults in other regions or in different countries remains uncertain. Fourth, although water intake may influence cellular health, this factor could not be accounted for in the present analysis.

5. Conclusion

In the present study, we found that PAL is an explanatory factor for the PhA and ECW/ICW ratio, as measured by the more accurate DLW method. Additionally, HLPa, even at a lower intensity than MVPA, was linked to the maintenance and/or improvement of the PhA and ECW/ICW ratio in older adults. These findings offer valuable insights for public health initiatives aimed at preventing sarcopenia in older adults in the future. However, further research should employ longitudinal study designs and/or larger sample sizes to allow for stratified analyses.

Glossary

BIA	Bioelectrical impedance analysis
BMI	Body mass index
BMR	Basal metabolic rate
DLW	Doubly labeled water
ECW	Extracellular water
EI	Energy intake
FFM	Fat-free mass
FFMI	Fat-free mass index
FFQ	Food frequency questionnaire
FM	Fat mass
HLPA	High-intensity light physical activity
IAEA	International Atomic Energy Agency
IBW	Ideal body weight
ICW	Intracellular water
LLPA	Low-intensity light physical activity
LPA	Light physical activity
MET	Metabolic equivalent of task
MVPA	Moderate-to-vigorous physical activity
PAL	Physical activity level
PBF	Percentage body fat
PhA	Phase angle
PI	Protein intake
SB	Sedentary behavior
SD	Standard deviation
TBW	Total body water
TEE	Total energy expenditure

CRediT authorship contribution statement

Kan Oishi: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Tsukasa Yoshida:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Data curation. **Hyeon-Ki Kim:** Writing – review & editing, Methodology, Conceptualization. **Keiichi Yokoyama:** Writing – review & editing, Investigation. **Yuya Watanabe:** Writing – review & editing, Investigation. **Yui Nakayama:** Writing – review & editing, Investigation. **Misaka Kimura:** Writing – review & editing, Project administration, Investigation, Funding acquisition. **Kojiro Ishii:** Writing – review & editing, Supervision. **Rei Ono:** Writing – review & editing, Supervision. **Yosuke Yamada:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Informed consent

All participants were informed of the experimental details and provided informed consent.

Ethical statement

This study was approved by the Ethics Committee of the Kyoto Prefectural University of Medicine (approval number: RBMR-E-363) and the National Institute of Health and Nutrition (NIHN187-3) and was conducted in compliance with the Declaration of Helsinki.

Funding statement

This work was supported by the Japan Society for the Promotion of Science (KAKENHI grant numbers: 24240091 to MK, 23K24782 to TY, and 24H00683 to YY). The funding source had no role in the study design; data collection, analysis, and interpretation; writing of the article; or decision to submit the article for publication.

Declaration of competing interest

The authors declare no conflicts of interest.

Acknowledgments

We sincerely thank all participants of the Kyoto–Kameoka study for their generous cooperation. We also acknowledge Editage (www.editage.com) for the professional English language editing of this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.exger.2025.112859>.

Data availability

The data that has been used is confidential.

References

- Akamatsu, Y., Kusakabe, T., Arai, H., Yamamoto, Y., Nakao, K., Ikeue, K., et al., 2022. Phase angle from bioelectrical impedance analysis is a useful indicator of muscle quality. *J. Cachexia. Sarcopenia Muscle* 13, 180–189. <https://doi.org/10.1002/jcsm.12860>.
- Amagasa, S., Machida, M., Fukushima, N., Kikuchi, H., Takamiya, T., Odagiri, Y., et al., 2018. Is objectively measured light-intensity physical activity associated with health outcomes after adjustment for moderate-to-vigorous physical activity in adults? A systematic review. *Int. J. Behav. Nutr. Phys. Act.* 15, 65. <https://doi.org/10.1186/s12966-018-0695-z>.
- Asano, Y., Tsuji, T., Okura, T., 2023a. Segmental extracellular-to-intracellular water resistance ratio and physical function in older adults. *Exp. Gerontol.* 181, 112278. <https://doi.org/10.1016/j.exger.2023.112278>.
- Asano, Y., Nagata, K., Shibuya, K., Fujii, Y., Kitano, N., Okura, T., 2023b. Association of 24-h movement behaviors with phase angle in community-dwelling older adults: a compositional data analysis. *Aging Clin. Exp. Res.* 35, 1469–1476. <https://doi.org/10.1007/s40520-023-02425-8>.
- Asano, Y., Tsuji, T., Kim, M., Nagata, K., Shibuya, K., Tateoka, K., et al., 2023c. Cross-sectional and longitudinal study of the relationship between phase angle and physical function in older adults. *Geriatr. Gerontol. Int.* 23 (2), 141–147. <https://doi.org/10.1111/ggi.14532>.
- Ayabe, M., Yahiro, T., Yoshioka, M., Higuchi, H., Higaki, Y., Tanaka, H., 2009. Objectively measured age-related changes in the intensity distribution of daily physical activity in adults. *J. Phys. Act. Health* 6, 419–425. <https://doi.org/10.1123/jpah.6.4.419>.
- Baumgartner, R.N., Chumlea, W.C., Roche, A.F., 1988. Bioelectric impedance phase angle and body composition. *Am. J. Clin. Nutr.* 48, 16–23. <https://doi.org/10.1093/ajcn/48.1.16>.
- Ceolin, J., de Borja, E.L., Mundstock, E., de Oliveira, J.R., Mattiello, R., Bodanese, L.C., 2023. Phase angle of bioimpedance as a marker of inflammation in cardiovascular diseases: a systematic review. *Nutrition* 112, 112064. <https://doi.org/10.1016/j.nut.2023.112064>.
- Costa Pereira, J.P.D., Rebouças, A.S., Prado, C.M., Gonzalez, M.C., Cabral, P.C., Diniz, A. D.S., et al., 2024. Phase angle as a marker of muscle quality: a systematic review and meta-analysis. *Clin. Nutr.* 43, 308–326. <https://doi.org/10.1016/j.clnu.2024.11.008>.
- da Costa Pedro, PereiraJ, Costa, J.H.S., de Miranda, G.B.L., de Sousa Rebouças, A., de Lima Bezerra, A.D., Lopes, M.M.G.D., et al., 2024. Prognostic value of overhydration and bioelectrical impedance vector on short- and long-term outcomes in hospitalized patients with cancer. *Clin. Nutr.* 43, 756–764. <https://doi.org/10.1016/j.clnu.2024.01.025>.
- Faul, F., Erdfelder, E., Buchner, A., Lang, A.G., 2009. Statistical power analyses using G*Power 3.1: tests for correlation and regression analyses. *Behav. Res. Methods* 41, 1149–1160. <https://doi.org/10.3758/BRM.41.4.1149>.
- Ganpule, A.A., Tanaka, S., Ishikawa-Takata, K., Tabata, I., 2007. Interindividual variability in sleeping metabolic rate in Japanese subjects. *Eur. J. Clin. Nutr.* 61 (11), 1256–1261. <https://doi.org/10.1038/sj.ejcn.1602645>.
- Garlini, L.M., Alves, F.D., Ceretta, L.B., Perry, I.S., Souza, G.C., Clausell, N.O., 2019. Phase angle and mortality: a systematic review. *Eur. J. Clin. Nutr.* 73, 495–508. <https://doi.org/10.1038/s41430-018-0159-1>.
- Haggarty, P., McGaw, B.A., Franklin, M.F., 1988. Measurement of fractionated water loss and CO₂ production using triply labelled water. *J. Theor. Biol.* 134, 291–308. [https://doi.org/10.1016/s0022-5193\(88\)80060-2](https://doi.org/10.1016/s0022-5193(88)80060-2).
- International Atomic Energy Agency, 2009. Assessment of Body Composition and Total Energy Expenditure in Humans Using Stable Isotope Techniques. In: IAEA Human Health Series No. 3. IAEA, Vienna, Austria. <https://www.iaea.org/publications/7982/assessment-of-body-composition-and-total-energy-expenditure-in-humans-using-stable-isotope-techniques> (accessed 19 November 2024).

- International Atomic Energy Agency, 2011. Introduction to Body Composition Assessment Using the Deuterium Dilution Technique with Analysis of Saliva Samples by Fourier Transform Infrared Spectrometry. In: IAEA Human Health Series No. 12. IAEA, Vienna, Austria. Available at: <https://www.iaea.org/publications/8369/introduction-to-body-composition-assessment-using-the-deuterium-dilution-technique-with-analysis-of-saliva-samples-by-fourier-transform-infrared-spectrometry> (accessed 19 November 2024).
- Iwasaka, C., Yamada, Y., Nishida, Y., Hara, M., Yasukata, J., Miyoshi, N., et al., 2023. Association of appendicular extracellular-to-intracellular water ratio with age, muscle strength, and physical activity in 8,018 community-dwelling middle-aged and older adults. *Arch. Gerontol. Geriatr.* 108, 104931. <https://doi.org/10.1016/j.archger.2023.104931>.
- Izawa, K.P., Shibata, A., Ishii, K., Miyawaki, R., Oka, K., 2017. Associations of low-intensity light physical activity with physical performance in community-dwelling elderly Japanese: a cross-sectional study. *PloS One* 12, e0178654. <https://doi.org/10.1371/journal.pone.0178654>.
- Kuchnia, A.J., Teigen, L.M., Cole, A.J., Mulasi, U., Gonzalez, M.C., Heymsfield, S.B., et al., 2017. Phase angle and impedance ratio: reference cut-points from the United States national health and nutrition examination survey 1999–2004 from bioimpedance spectroscopy data. *Journal of Parenteral and Enteral Nutrition* 41, 1310–1315. <https://doi.org/10.1177/0148607116670378>.
- Lukaski, H.C., 1996. Biological indexes considered in the derivation of the bioelectrical impedance analysis. *Am. J. Clin. Nutr.* 64, 397S–404S. <https://doi.org/10.1093/ajcn/64.3.397S>.
- Lukaski, H.C., Kyle, U.G., Kondrup, J., 2017. Assessment of adult malnutrition and prognosis with bioelectrical impedance analysis: phase angle and impedance ratio. *Curr. Opin. Clin. Nutr. Metab. Care* 20, 330–339. <https://doi.org/10.1097/MCO.0000000000000387>.
- Mattiello, R., Amaral, M.A., Mundstock, E., Ziegelmann, P.K., 2020. Reference values for the phase angle of the electrical bioimpedance: systematic review and meta-analysis involving more than 250,000 subjects. *Clin. Nutr.* 39, 1411–1417. <https://doi.org/10.1016/j.clnu.2019.07.004>.
- Müller, E.A., 1970. Influence of training and of inactivity on muscle strength. *Arch. Phys. Med. Rehabil.* 51, 449–462.
- Nakashima, D., Fujii, K., Tsubouchi, Y., Kubo, Y., Yorozuya, K., Noritake, K., et al., 2024. Relationship between phase angle and physical activity intensity among community-dwelling older adults in Japan: a cross-sectional study. *Healthcare (Basel)* 12, 167. <https://doi.org/10.3390/healthcare12020167>.
- Norman, K., Stobäus, N., Pirlich, M., Bösby-Westphal, A., 2012. Bioelectrical phase angle and impedance vector analysis—clinical relevance and applicability of impedance parameters. *Clin. Nutr.* 31, 854–861. <https://doi.org/10.1016/j.clnu.2012.05.008>.
- Ohashi, Y., Joki, N., Yamazaki, K., Kawamura, T., Tai, R., Oguchi, H., et al., 2018. Changes in the fluid volume balance between intra- and extracellular water in a sample of Japanese adults aged 15–88 yr old: a cross-sectional study. *Am. J. Physiol. Ren. Physiol.* 314, F614–F622. <https://doi.org/10.1152/ajprenal.00477.2017>.
- Osuka, Y., Yabushita, N., Kim, M., Seino, S., Nemoto, M., Jung, S., et al., 2015. Association between habitual light-intensity physical activity and lower-extremity performance: a cross-sectional study of community-dwelling older Japanese adults. *Geriatr. Gerontol. Int.* 15, 268–275. <https://doi.org/10.1111/ggi.12268>.
- Pollock, M.L., Gaesser, G.A., Butcher, J.D., Després, J.P., Dishman, R.K., Franklin, B.A., et al., 1998. American college of sports medicine position stand: the recommended quantity and quality of exercise for developing and maintaining cardiorespiratory and muscular fitness, and flexibility in healthy adults. *Med. Sci. Sports Exerc.* 30, 975–991. <https://doi.org/10.1097/00005768-199806000-00032>.
- Racette, S.B., Schoeller, D.A., Luke, A.H., Shay, K., Hnilicka, J., Kushner, R.F., 1994. Relative dilution spaces of 2H- and 18O-labeled water in humans. *Am. J. Physiol.* 267, E585–E590. <https://doi.org/10.1152/ajpendo.1994.267.4.E585>.
- Ramsey, K.A., Rojer, A.G.M., D'Andrea, L., Otten, R.H.J., Heymans, M.W., Trappenburg, M.C., et al., 2021. The association of objectively measured physical activity and sedentary behavior with skeletal muscle strength and muscle power in older adults: a systematic review and meta-analysis. *Ageing Res. Rev.* 67, 101266. <https://doi.org/10.1016/j.arr.2021.101266>.
- Sagayama, H., Yamada, Y., Racine, N.M., Shriver, T.C., Schoeller, D.A., DLW Study Group, 2016. Dilution space ratio of 2H and 18O of doubly labeled water method in humans. *J. Appl. Physiol.* 120 (11), 1349–1354. <https://doi.org/10.1152/japplphysiol.01037.2015>.
- Sato, H., Nakamura, H., Nishida, Y., Shirahata, T., Yogi, S., Akagami, T., et al., 2021. Energy expenditure and physical activity in COPD by doubly labelled water method and an accelerometer. *ERJ Open Res.* 7, 00407–2020. <https://doi.org/10.1183/23120541.00407-2020>.
- Speakman, J.R., Yamada, Y., Sagayama, H., Berman, E.S.F., Ainslie, P.N., Andersen, L.F., et al., 2021. A standard calculation methodology for human doubly labeled water studies. *Cell Rep Med.* 2, 100203. <https://doi.org/10.1016/j.xcrm.2021.100203>.
- Watanabe, D., Nanri, H., Sagayama, H., Yoshida, T., Itoi, A., Yamaguchi, M., et al., 2019a. Estimation of energy intake by a food frequency questionnaire: calibration and validation with the doubly labeled water method in Japanese older people. *Nutrients* 11, 1546. <https://doi.org/10.3390/nu11071546>.
- Watanabe, D., Nanri, H., Yoshida, T., Yamaguchi, M., Sugita, M., Nozawa, Y., et al., 2019b. Validation of energy and nutrition intake in Japanese elderly individuals estimated based on a short food frequency questionnaire compared against a 7-day dietary record: the Kyoto-Kameoka study. *Nutrients* 11 (3), 688. <https://doi.org/10.3390/nu11030688>.
- Watanabe, D., Kuratani, K., Yoshida, T., Nanri, H., Watanabe, Y., Date, H., et al., 2022. Diet quality and physical or comprehensive frailty among older adults. *Eur. J. Nutr.* 61, 2451–2462. <https://doi.org/10.1007/s00394-022-02819-w>.
- Watanabe, D., Yoshida, T., Watanabe, Y., Kimura, M., Yamada, Y., Kyoto-Kameoka Study Group, 2023. Doubly labelled water-calibrated energy intake associations with mortality risk among older adults. *J. Cachexia Sarcopenia Muscle* 14, 214–225. <https://doi.org/10.1002/jcsm.13122>.
- Weir, J.B., 1949. New methods for calculating metabolic rate with special reference to protein metabolism. *J. Physiol.* 109, 1–9. <https://doi.org/10.1113/jphysiol.1949.sp004363>.
- Westerterp, K.R., 1999. Assessment of physical activity level in relation to obesity: current evidence and research issues. *Med. Sci. Sports Exerc.* 31, S522–S525. <https://doi.org/10.1097/00005768-199911001-00006>.
- Yamada, Y., Schoeller, D.A., Nakamura, E., Morimoto, T., Kimura, M., Oda, S., 2010. Extracellular water may mask actual muscle atrophy during aging. *J. Gerontol. A Biol. Sci. Med. Sci.* 65, 510–516. <https://doi.org/10.1093/gerona/glq001>.
- Yamada, Y., Yoshida, T., Yokoyama, K., Watanabe, Y., Miyake, M., Yamagata, E., et al., 2017a. The extracellular to intracellular water ratio in upper legs is negatively associated with skeletal muscle strength and gait speed in older people. *J. Gerontol. A Biol. Sci. Med. Sci.* 72, 293–298. <https://doi.org/10.1093/gerona/glw125>.
- Yamada, Y., Nanri, H., Watanabe, Y., Yoshida, T., Yokoyama, K., Itoi, A., et al., 2017b. Prevalence of frailty assessed by Fried and Kihon checklist indexes in a prospective cohort study: design and demographics of the Kyoto-Kameoka longitudinal study. *J. Am. Med. Dir. Assoc.* 18, 733.e7–733.e15. <https://doi.org/10.1016/j.jamda.2017.02.022>.
- Yamada, Y., Hashii-Arishima, Y., Yokoyama, K., Itoi, A., Adachi, T., Kimura, M., 2018. Validity of a triaxial accelerometer and simplified physical activity record in older adults aged 64–96 years: a doubly labeled water study. *Eur. J. Appl. Physiol.* 118, 2133–2146. <https://doi.org/10.1007/s00421-018-3944-6>.
- Yamada, Y., Yoshida, T., Murakami, H., Kawakami, R., Gando, Y., Ohno, H., et al., 2022. Phase angle obtained via bioelectrical impedance analysis and objectively measured physical activity or exercise habits. *Sci. Rep.* 12, 17274. <https://doi.org/10.1038/s41598-022-21095-6>.
- Yamada, Y., Yoshida, T., Murakami, H., Gando, Y., Kawakami, R., Ohno, H., et al., 2023. Body cell mass to fat-free mass ratio and extra- to intracellular water ratio are related to maximal oxygen uptake. *J. Gerontol. A Biol. Sci. Med. Sci.* 78, 1778–1784. <https://doi.org/10.1093/gerona/glad140>.